

Dear Editors:

We would like to submit the enclosed manuscript entitled **“The Pipeline Overshoots the Shock: An Adaptive-Belief Systems Model of Delayed Educational Supply and Graduate Labour-Market Adjustment”**, which we wish to be considered for publication in *Systems* for the Special Issue **“Economic System Management, Sustainability, and Innovation in Digital Environments.”**

No conflict of interest exists in the submission of this manuscript, and the manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously and is not under consideration for publication elsewhere, in whole or in part. All authors listed have approved the manuscript that is enclosed.

A degree takes four years to produce, so the students who react to a labour market reach it four years late. This study asks what that delay does to an education system when demand for one field falls abruptly, as demand for entry-level computing work has since 2023. We build a two-field model in which firms hire entry-tier graduates through a matching market with free entry, the two kinds of graduate are imperfect substitutes in production so that a crowded field is one in which graduates are both poorly paid and hard to place, and entrants choose a field by a discrete-choice rule after forecasting what it will be worth on graduation. Forecasts come from two competing rules—one anchored on the field’s long-run value, one that projects the recent trend—and the weight on each evolves with its recent accuracy, which makes the education system an adaptive belief system in the sense of Brock and Hommes. The labour block is calibrated to four moments of the pre-shock graduate market and matches them to thirteen decimal places; the size of the technology shock is then set so that the model reproduces the new-graduate unemployment rate now observed in the exposed field, which leaves the entire transition free to be right or wrong. The model is solved, calibrated and simulated from a single seeded pipeline, and it uses no individual-level data: every calibration target is a published aggregate.

The innovations of this paper are as follows.

(1) This study brings adaptive belief systems to the education market, where the production lag is a degree. The Brock–Hommes mechanism—agents switching between forecasting rules according to recent accuracy, with the intensity of switching as the bifurcation parameter—has been applied to asset markets, to macroeconomic expectations and to commodity markets with production lags. We could not find it applied to the choice of what to study, which is the market with the longest production lag of all and the one in which a bad forecast is least reversible. Embedding it in an equilibrium search-and-matching labour block, rather than in a reduced-form demand curve, is what allows the model to speak about unemployment as well as wages, and it is what lets the current configuration of the exposed field—a high early-career wage alongside an unusually high new-graduate unemployment rate—be reproduced as a property of the transition rather than assumed.

(2) The enrolment response overreacts by a factor of two, and the delay alone does not explain it. The entering share falls 28.5 per cent on impact when the warranted long-run fall is 13.9 per cent. Most of the enrolment collapse that follows such a shock is therefore transitional rather than warranted. The decomposition across belief regimes is the part we regard as most informative: with beliefs anchored on the field's long-run value the amplification is only 1.10, so a four-year delay by itself produces very nearly the right answer. It is the competition between forecasting rules that does the work, and the amplification rises monotonically to 1.89 when a trend-projecting rule is present and to 2.48 when entrants imitate the recently successful rule sharply. The result survives a global sensitivity sweep: the amplification exceeds one in every draw that admits an exact recalibration.

(3) The size of a system's correction is not a measure of how well it is behaving, and we show that an indicator built on it ranks interventions backwards. This is the finding we did not expect and the one with the clearest practical content. Making the signal about current returns noisier shrinks the enrolment swing markedly, from an amplification of 2.05 to 1.65, and it raises the adjustment cost by 14 per cent. Publishing the enrolment pipeline—the cohorts already enrolled and not yet graduated—does the opposite: it makes the immediate response deeper and lowers the adjustment cost by about six per cent. The two rankings are opposite because the deeper response is the better-informed one: at the moment the shock lands, four large pre-shock cohorts are still coming, and an entrant who can see them correctly infers that the field will stay crowded. A ministry that judged an information policy by whether it calmed enrolment swings would therefore reject the policy that improves allocation and adopt the one that degrades it. The stabilising series is an administrative by-product that education systems already hold and almost none publish in usable form.

We believe that this manuscript fits well within the scope of *Systems* and the Special Issue “Economic System Management, Sustainability, and Innovation in Digital Environments”. Its subject is the management of an economic system under a digital-technology shock: how a coupled education–labour system allocates people between fields when the information it runs on arrives later than the decisions it governs. The paper is explicitly systemic in its machinery—state variables, a production delay, a balancing loop crossed with a reinforcing one, stability analysis of the stationary state and a global sensitivity sweep—and it is quantitative economics throughout, which we hope complements the more empirical and management-oriented contributions the Special Issue has attracted. Its policy content concerns resource allocation and the design of information flows under digital transformation, which sit squarely in the call's keyword list.

We deeply appreciate your consideration of our manuscript and look forward to receiving comments from the reviewers. Correspondence should be directed to Xinyu Cai at the following address:

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Thanks very much for your attention to our paper.

Very sincerely yours,

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